

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/380746219>

Prevalence of opportunistic pathogens and anti-microbial resistance in urban aquaculture ponds

Article in *Journal of Hazardous Materials* · May 2024

DOI: 10.1016/j.jhazmat.2024.134661

CITATIONS

4

READS

111

11 authors, including:



Javed Foysal

Shahjalal University of Science and Technology

112 PUBLICATIONS 1,956 CITATIONS

SEE PROFILE



A.Q.M. Robiul Kawser

Gazipur Agricultural University

17 PUBLICATIONS 198 CITATIONS

SEE PROFILE



Md Reaz Chaklader

Department of Primary Industries and Regional Development

76 PUBLICATIONS 1,789 CITATIONS

SEE PROFILE



Prevalence of opportunistic pathogens and anti-microbial resistance in urban aquaculture ponds

Md Javed Foysal^{a,b,c,*}, A Q M Robiul Kawser^{d,e}, Sulav Indra Paul^{f,g}, Md Reaz Chaklader^h, Sanjay Kumar Guptaⁱ, Alfred Tay^j, Brett A. Neilan^a, Marthe Monique Gagnon^b, Ravi Fotedar^b, Md Mahbubur Rahman^f, Verlaine J. Timms^a

^a School of Environmental and Life Sciences, The University of Newcastle, Callaghan, NSW, Australia

^b School of Molecular and Life Sciences, Curtin University, Perth, WA, Australia

^c Department of Genetic Engineering and Biotechnology, Shahjalal University of Science and Technology, Sylhet, Bangladesh

^d Department of Aquaculture, Bangabandhu Sheikh Mujibur Rahman Agricultural University, Gazipur, Bangladesh

^e School of Veterinary Medicine and Science, University of Nottingham, United Kingdom

^f Institute of Biotechnology and Genetic Engineering, Bangabandhu Sheikh Mujibur Rahman Agricultural University, Gazipur, Bangladesh

^g Institute for Biosecurity and Microbial Forensics, Oklahoma State University, OK, USA

^h Department of Primary Industries and Regional Development, Fremantle, WA, Australia

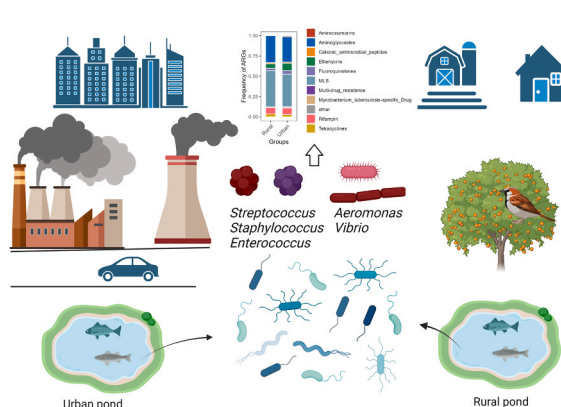
ⁱ ICAR-Indian Institute of Agricultural Biotechnology, Ranchi, Jharkhand, India

^j School of Biomedical Sciences, University of Western Australia, Perth, Australia

HIGHLIGHTS

- Bacterial diversity and antimicrobial resistance were investigated in rural and urban aquaculture ponds.
- Higher bacterial diversity and more opportunistic pathogens were observed in urban ponds.
- Metal resistance classes were observed only in urban ponds.
- Beta and gamma-proteobacteria were involved in the majority of resistance.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Opportunistic pathogens
Metal resistance
Bacterial interactions
Environmental management and sustainability

ABSTRACT

Bacterial antimicrobial resistance (AMR) has emerged as a significant concern worldwide. The microbial community profile and potential AMR level in aquaculture ponds are often undervalued and attract less attention than other aquatic environments. We used amplicon and metagenomic shotgun sequencing to study microbial communities and AMR in six freshwater polyculture ponds in rural and urban areas of Bangladesh. Amplicon sequencing revealed different community structures between rural and urban ponds, with urban ponds having a

* Correspondence to: School of Environmental and Life Sciences, The University of Newcastle, Callaghan, NSW 2308, Australia.

E-mail addresses: javed.foysal@gmail.com, mdjaved.foysal@newcastle.edu.au (M.J. Foysal).

<https://doi.org/10.1016/j.jhazmat.2024.134661>

Received 13 October 2023; Received in revised form 15 May 2024; Accepted 18 May 2024

Available online 20 May 2024

0304-3894/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

higher bacterial diversity and opportunistic pathogens including *Streptococcus*, *Staphylococcus*, and *Corynebacterium*. Despite proteobacterial dominance, Firmicutes was the most interactive in the community network, especially in the urban ponds. Metagenomes showed that drug resistance was the most common type of AMR found, while metal resistance was only observed in urban ponds. AMR and metal resistance genes were found mainly in beta and gamma-proteobacteria in urban ponds, while AMR was found primarily in alpha-proteobacteria in rural ponds. We identified potential pathogens with a high profile of AMR and metal resistance in urban aquaculture ponds. As these ponds provide a significant source of protein for humans, our results raise significant concerns for the environmental sustainability of this food source and the dissemination of AMR into the food chain.

1. Introduction

Aquaculture has evolved from a supplemental food source to a critical part of global food security and aquatic protein production systems [1]. Bangladesh, a country synonymous with lush waters and rich aquatic heritage, has become a key participant in this transformation [2, 3]. The twin challenges of feeding a rapidly growing global population and providing livelihood opportunities in light of dwindling commercial fishing have fueled the expansion of aquaculture in the country's rural and urban landscapes [4]. Polyculture, an integrated approach involving the simultaneous farming of multiple species in the same body of water, has already gained significant momentum as an efficient method to optimize productivity and resource use in aquaculture ponds [5]. In this type of pond, the structural and interactive intricacies of the microbes play a critical role in shaping the ecosystem dynamics [6].

Microbial communities including bacteria, archaea, viruses, and fungi, establish complex inter-dependencies. These interdependencies directly affect nutrient cycling, water quality, disease dynamics and biogeochemical cycles [7], providing stability to aquatic environments [8]. Such multispecies microbial communities in a distinct habitat are commonly defined as 'microbiome' [9]. The heterogeneity of the microbiome in different regions of polyculture ponds is influenced by a variety of anthropogenic factors. These include chemical supplements, feed inputs, management practices, and both industrial and medical effluents [6,10,11]. Bacteria play a crucial role as the primary drivers of nutrient recycling, organic matter decomposition, and waste remediation in the environmental microbiome [12]. They significantly influence water quality parameters such as dissolved oxygen, ammonia, and nitrogen metabolites [13]. A shift in the structure and composition of bacteria can lead to changes in resistance profiles and functional capabilities, ultimately affecting both environmental and human health [14]. Constant aquatic pollution and the use of chemicals have contributed to the spread of antibiotic-resistant bacteria (ARB) and antimicrobial resistance (AMR) genes in different water bodies including aquaculture ponds [15–19]. Aquatic organisms, including plankton, can uptake ARB from rearing water and transfer them to humans through the food chain [20,21]. Furthermore, previous studies have revealed the transmission of pathogenic bacteria to aquatic environments, with *Escherichia-Shigella*, *Salmonella*, and *Vibrio* being detected in freshwater sources [22–25]. Despite reports of widespread aquatic pollution, no study has yet compared the microbial communities and prevalence of ARB and AMR between rural and urban aquaculture ponds in developing countries to understand their impacts on human and environmental health.

The emergence and spread of AMR and metal resistance genes (MRG) in the last two decades have raised serious concerns for human and animal health, food safety and environmental sustainability [26]. The proliferation of antimicrobial agents in aquaculture systems primarily results from domestic and agricultural waste, contaminated feed ingredients, various industrial effluents, and extensive use of antibiotics [27–29]. AMR and MRG can also be incorporated into aquatic habitats through horizontal gene transfer by mobile genetic elements [28,30]. These resistance genes can then circulate and become enriched in ponds through host-microbe or microbe-metal interactions [31]. A wide range of AMR and MRG have also negatively impacted the growth and

immunity of aquatic animals. Given that aquaculture ponds provide major sources of income and nutrition for low- and middle-income people in Bangladesh, a study on the distribution and dissemination of AMR and MRG is thus a pressing priority. In addition, it has been established that environmental factors are directly linked to 90% of human diseases [32]. Lifetime exposure to microbial and environmental hazards including AMR, is referred to as 'eco-exposome' and significantly influences human health and immunity [33,34]. Investigating the dynamic relationship between the environmental microbiome and resistome profile with the eco-exposome (which includes pharmaceuticals, antibiotics, and heavy metals) and incorporating this information into a public health database, would support strategies for controlling the exposome and protecting public health.

The environment harbors a rich reserve of microbial communities, but their identification can be challenging. A decade ago, most isolation and identification methods relied on culture-dependent approaches [35] however, over 99% of environmental microbes are non-culturable and remain undetectable through traditional methods [36]. Consequently, culture-independent high-throughput sequencing has gained more popularity in microbiome studies due to its ability to detect numerous species and their genomes with higher accuracy [37,38]. Amplicon sequencing, often referred to as 16S rRNA gene sequencing or metabarcoding, has been widely used to study microbial diversity and composition [39]. While 16S rRNA sequencing is targeted and cost-effective, it cannot resolve full metagenomes and has low sensitivity at the species level [40]. To understand the genomic features and functional capabilities of microbes, such as toxin genes, secondary metabolites, and AMR, shotgun metagenomics is necessary [41]. A comprehensive study that combines amplicon and metagenome sequencing is essential to gain insights into the microbiome and identify any potential threats to environmental sustainability and human health. Therefore, the primary objectives of this study were to (i) characterize the microbiome in rural and urban polyculture ponds using amplicon sequencing; and (ii) assess the prevalence and distribution of host-associated AMR and MRG through metagenome sequencing. We hypothesized that urban aquaculture ponds might provide a potential environment for AMR due to aquatic pollution. The results of this study will facilitate the development of suitable pond management strategies to mitigate the risks and threats.

2. Methods

2.1. Sample collection and processing

Water samples were collected from aquaculture ponds in two different regions of Bangladesh. Six polyculture ponds, three from each site of Salna, Gazipur (24.0290° N, 90.3862° E) and Adamdighi, Bogra (24.8150° N 89.0104° E) were selected for sampling. The ponds were used to farm Indian major carp (*Labeo rohita*, *Catla catla*, and *Cirrhinus cirrhosis*) for at least five years with a density of 40 fish/decimal (1 fish/m²). The urban site (Salna) is located 25 km away from the capital, Dhaka, and is a densely populated area surrounded by garment, textile, pharmaceutical, and chemical industries. In contrast, the second site is 275 km away from Dhaka, has no major industrial and human activities

within a 25 km radius, and is considered a rural site (Fig. S1). Neither site is protected by any national or international regulations including codex regulations. The research was monitored by the Dean of Graduate Research, Institute of Genetic Engineering and Biotechnology, Salna, Gazipur, Bangladesh.

Apart from regional differences, the ponds were similar in terms of their age (15 ± 5 years), size ($8061 \pm 36 \text{ m}^2$ equivalent to 0.8-hectare land), cultured species, and the nutrition profile of feeding pellets. Both sites' ponds were replenished with groundwater and rainwater sources every fortnight, depending on the seasons and intensity of rainfall. During periods of heavy rainfall, these ponds occasionally receive input from surface runoff from the surrounding areas. Annually, the ponds were completely drained, dried out, limed and disinfected using bleaching powder before restocking. For growth and nutrition, fish were fed a fishmeal-based diet (Feed A, Fish Feed Extruder, Dhaka, Bangladesh) which consists of a combination of protein, lipid, ash and fiber. Further details regarding feeding ingredients and the proximate composition of diet have been described in our previously published article [42]. Water samples were collected in 500 mL sterile bottles from four different corners of the ponds at a depth of 1 m during the rainy season (July–August). The collected water samples were immediately transported to the laboratory in chilled condition for further analysis. Water samples (1 L) were filtered using $0.2 \mu\text{M}$ glass filters which were stored at -30°C until further processing.

2.2. Water quality

Dissolved oxygen (DO) concentration was measured using a portable DO meter (Socialme, China). The temperature and pH of the water were monitored using a portable waterproof $^\circ\text{C}/\text{mV}/\text{pH}$ meter (CyberScan pH 300, Eutech Instruments, Singapore). Additionally, the concentrations of nitrate and nitrite were measured using a colorimeter (Hach, CO, USA).

2.3. DNA extraction and PCR amplification

Genomic DNA extraction was performed using the FastDNA Spin Kit for Soil (MP Biomedicals, CA, USA). DNA concentration was measured with a nanodrop spectrophotometer 2000cc (Thermo Fisher Scientific, USA) and Qubit 4 (Invitrogen Life Sciences, USA). For metagenomic sequencing, DNA from three respective ponds of rural and urban samples were pooled into a final volume of $150 \mu\text{L}$ and purified using the Power Clean (Qiagen, Hilden, Germany) clean-up kit. The quality was assessed using a 1% agarose gel to check for any shredding or fragmenting of DNA. For amplicon sequencing, PCR amplification was performed with V3V4 hypervariable region primers [43] containing Illumina adapter sequences. Briefly, a $50 \mu\text{L}$ master mix was prepared, including $2 \mu\text{L}$ of 30 ng template DNA, $1 \mu\text{L}$ of $10 \mu\text{M}$ each primer, $25 \mu\text{L}$ of a prepared mix (Hot Start 2X, New England BioLabs Inc., USA), and $21 \mu\text{L}$ of DEPC-treated water. The PCR amplification protocol consisted of an initial denaturation at 95°C for 5 min, denaturation at 95°C for 30 s, annealing at 55°C for 40 s, extension at 72°C for 90 s and final extension at 72°C for 10 min. A Bio-Rad S1000 Thermal Cycler (BioRad Laboratories Inc., USA) was used for 30 cycles of PCR amplification.

2.4. Amplicon and metagenome sequencing

Positive amplicons were purified using AMPure beads, normalized to an equimolar concentration, and indexed with the Nextera XT Kit following Illumina 16S Sequencing Library Preparation (Part # 15044223 Rev. B). The final set of equimolar pooled amplicons was sequenced on an Illumina MiSeq platform (Illumina Inc., CA, USA) using a v3 kit (300 bp and 600 cycles). For metagenome sequencing, purified DNA was sequenced in a SP lane (150 bp, 100 Gbp) using NovaSeq platform (Illumina Inc., CA, USA).

2.5. Amplicon data processing and analysis

Paired-end raw data were imported and processed in qiime2 (v2021.11) [44]. The denoising process involved quality trimming of demultiplexed reads, removing chimeras and singletons, and clustering of sequences into amplicon sequence variants (ASVs). Denoising was performed using the DADA2 plug-in in qiime2 [45]. The ASV table was further filtered at the lowest non-zero frequency (10) and classified against SILVA 138 release using a consensus blast classifier [46]. To reduce the risk of crossing over, the feature table was collapsed with taxonomy data and exported as a TSV file. Chloroplast and mitochondrial DNA sequences were removed from the final dataset. For consistency throughout the data analysis, "*Clostridium sensu stricto* 1–9" and "*Prevotella* 1–9" were renamed as *Clostridium* and *Prevotella*, respectively. Unclassified reads, including uncultured, ambiguous taxa, and unknown were grouped as "others". Considering the number of genera obtained, phylum level was considered for the network analysis of the bacterial community. The network was constructed based on ASV similarities in all samples between rural and urban ponds. The connection between edges and linkages was visualized using R packages networkD3 and igraph, resulting in a circular visualization plot [47,48]. Strong and significant interactions were determined based on a Spearman correlation value of $R \geq 0.7$ and p -value of < 0.01 . The final network was visualized and edited using Gephi software [49].

2.6. Metagenomic assembly, binning and read classifications

Low-quality bases were trimmed using bbdutk (v39.1) with the following parameters: trimq = 20, minq = 11 [50]. The quality of raw and trimmed reads was assessed using FastQC software (v0.12.1) [51]. Metagenomic reads were assembled using Megahit (v1.2.9) with default parameters [52]. Following assembly, contigs shorter than 1000 bp were removed to avoid ambiguous contig annotation. To map the reads back to contigs, we employed the Bowtie2 alignment mapper [53]. The resulting SAM files were converted to BAM format using SAMtools (v1.17) [54]. Metagenomic binning was performed with MetaBAT2 (v2.15) and MaxBin2 (v2.2.7) [55,56]. The bins were further purified with MAGpurify (v2.1.2) [57] and dereplicated using DAS tools (v1.1.6) [58]. The quality, coverage and contamination of each bin were checked using CheckM (v1.2.2) [59]. Only high-quality bins meeting the criteria of > 90 completeness and $< 10\%$ contamination or medium-quality bins with > 60 completeness and $< 20\%$ contamination were selected as metagenome-assembled genome (MAGs) for further analysis. These selected bins were taxonomically classified against the Kraken2 microbial database [60].

2.7. Screening of antimicrobial and metal resistance genes

AMR+ + in conjunction with MEGARes (v3.0) was employed to screen raw metagenomic datasets for antimicrobial and metal resistance using the Nextflow standard_AMR pipeline with a conda profile [61]. Briefly, the pipeline involved trimming, host removal, resistome alignment, and hit classification for AMR and MRG. The obtained data were classified based on type and classes of antimicrobial and metal resistance.

2.8. Screening of bin-specific antimicrobial resistance and toxins

High- and medium-quality bins were screened for the presence of antimicrobial and metal resistance against Kraken2-classified selected bacterial phyla using abritAMR [62]. Additionally, we curated representative sequences from UniProtKB (<https://www.uniprot.org/>) for extended-spectrum beta-lactamase (ESBL)-producing genes from clinical isolates, metal-resistant genes from the environment and microcystin (*mcy*) toxin gene clusters from *Microcystis* sp. Subsequently, a customized database was created using diamond make-db, which was

used to run BLAST against high- and medium-quality bins with a similarity threshold >50 % and an error rate of $e10$ [63]. A BLAST identity score above >95 % and 90 % coverage was considered full gene recovery. The BLAST results for *mcy* toxin production were then validated using antiSMASH. A phylogenomic study of high and medium quality bins was performed using GToTree (<https://github.com/AstroBioMik/e/GToTree>). The tree was visualized and annotated using iTOL [64].

2.9. Statistical analysis

All statistical analyses were performed in R software. The Student's *t*-test was used to compare alpha diversity between urban and rural samples. For beta-diversity analysis, we conducted Principal Coordinate Analysis (PCoA), and Permutational Multivariate Analysis of Variance (PERMANOVA) was applied to compare clustering ordination using *anodis2* with 999 permutations. The PERMANOVA *R* and *p*-value were calculated for both unweighted and weighted UniFrac distance metrics in the *vegan* R package. Additionally, differential abundance analysis of genera between rural and urban samples was performed as Linear Discriminant Analysis (LDA) with an LDA value of ≥ 2.0 using the MicrobiomeMarker R package [65].

3. Results

3.1. Water quality

Nitrate and nitrite concentrations were significantly higher in the urban ponds than in the rural ponds. However, dissolved oxygen concentration was found lower in the urban ponds (Fig. S2). No difference in water temperature (24.8 ± 1.4 °C) was observed at a depth of 10 cm between rural and urban ponds.

3.2. Sequence quality and diversity metrics

In total, 531,527 high-quality V3V4 amplicon reads were obtained from 24 samples, with an average of $22,146.9 \pm 4689.4$ reads per sample. The lowest depth was 13,570 while the highest reached 36,308. Reads were assigned to 1503 ASVs (1058 ASVs following rarefaction), 25 phyla, 112 orders, and 286 genera. The top 10 ASVs covered 72% of reads across the 24 samples. The rarefied data and rarefaction curve revealed that each sample was sequenced at adequate depth to cover maximum diversity and composition (Fig. S3). The number of observed ASVs was significantly higher in urban samples. Furthermore, the most abundant ASVs were more evenly distributed across the urban samples, as reflected by higher Shannon and Simpson indices (Fig. 1A). More than twice as many unique ASVs were detected in urban samples (663) compared to rural ponds (309), with only 86 ASVs shared (Fig. 1B). The rural and urban bacterial communities were distinctly separated in the beta-diversity PCoA plot based on Bray-Curtis dissimilarity. The PERMANOVA *R*-value ($R^2 = 0.1506$) and *p*-value ($p < 0.001$) indicated significant structural and compositional variations in bacterial communities (Fig. 1C).

3.3. Amplicon-based bacterial composition

The bacterial communities in the studied ponds exhibit distinct patterns. Proteobacteria dominate (>50%) across all samples, followed by Fusobacteria and Actinobacteria. Notably, Firmicutes (18%) and Bacteroidota (5%) were significantly more abundant in urban ponds compared to rural samples. Meanwhile, Planctomycetota (5%) and Verrucomicrobia (1%) were abundant only in rural samples (Fig. 1D). At the genus level, several key bacteria play a crucial role in both rural and urban ponds. These include *Aeromicrobium*, *Aeromonas*, *Cetobacterium* and *Escherichia-Shigella*, which served as 'core' water bacteria for Indian major carp aquaculture. Additionally, *Vibrio* and *Lactococcus* were detected exclusively in rural ponds (Fig. 1E; Fig. S4). Urban ponds

harbored eight unique genera – *Cutibacterium*, *Pseudomonas*, *Sphingobacterium*, *Staphylococcus*, *Streptococcus*, *Enterococcus*, *Neisseria*, *Corynebacterium* and *Faecalibacterium*. Among these, *Pseudomonas* (28.5%), *Cetobacterium* (13%), and *Escherichia-Shigella* (15.5%) comprised 57% of total read abundance in urban ponds. Furthermore, clinical pathogens such as *Staphylococcus* (7.3%), *Streptococcus* (2.7%), *Corynebacterium* (1.6%) were also detected in all urban ponds (Fig. 1F). In contrast, 73.6% of the reads from rural ponds were classified as *Aeromicrobium* (13.5%), *Aeromonas* (18.1%), *Cetobacterium* (16.4%), *Escherichia-Shigella* (18.2%), and *Vibrio* (7.4%). *Cetobacterium* and *Escherichia-Shigella* emerged as the main core bacteria in both rural and urban ponds, exhibiting high and uniform abundance across all samples (Fig. 1G). Firmicutes were found to play a central role, showing strong interactions with Actinobacteria and Proteobacteria in the Spearman correlation network. Interestingly, despite being highly abundant, Proteobacteria was less interactive in the correlation network (Fig. 2). In total, 46 genera showed significantly higher abundance – 29 in urban and 17 in rural ponds. Notably, potential pathogens including *Staphylococcus*, *Streptococcus*, *Pseudomonas*, *Faecalibacterium*, *Neisseria* and *Corynebacterium* showed significantly higher abundance in urban ponds (Fig. 3).

3.4. Metagenome-based bacterial composition

Metagenomic assembly yielded 250 contigs, 186 of them above 1000 bp in length. Read-based classification showed a high abundance of Proteobacteria, specifically gamma-proteobacteria in all samples. Similar to amplicon data, *Aeromicrobium*, *Aeromonas*, *Cetobacterium* and *Vibrio* were dominant genera in rural samples. *Cyanobium*, *Limnohabibans*, *Shewanella* and *Vogesella* were plentiful in all samples despite being absent in the amplicon data. *Escherichia coli* abundance was lower in the rural ponds than in urban ones. *Enterococcus*, *Micrococcus*, *Staphylococcus*, and *Serratia* were identified from the metagenomic dataset for urban samples. In addition to bacteria, archaea, specifically ammonia oxidizing *Nitrososphaera* was detected in both rural and urban ponds (Fig. S5).

3.5. Antimicrobial and metal resistance genes

A total of 132 resistance subtypes were detected from the metagenome dataset. The set includes 17 major AMR and MRG classes – aminocoumarins, aminoglycosides, cationic antimicrobial peptides, fluoroquinolones, *Mycobacterium tuberculosis*-specific drug, MLS (macrolides, lincosamides, and streptogramins), multidrug resistance, rifampin, tetracyclines, elfamycins, copper resistance, drug and biocide resistance, mercury resistance, iron resistance, oxazolidinone, aminocoumarins, arsenic and peroxide resistance. In total, 45 AMR and MRG hits were found in the rural ponds compared to 84 in the urban ones. Nearly all antibiotic resistance in rural ponds was classified as drug (antibiotic) resistance subclass. The urban ponds displayed a mixture of drug and metal resistance subclasses (Fig. 4A). All 17 AMR and MRG subclasses were detected in the urban ponds, compared to only nine in rural samples. Out of these nine, subclass aminoglycosides and MLS contributed more than 60% of the drug resistance in rural ponds, compared to 40% in urban ponds. Eight unique AMR and MRG subclasses in urban ponds were aminocoumarins, arsenic resistance, copper resistance, drug and biocide resistance, iron resistance, mercury resistance, oxazolidinone, and peroxide resistance. Copper, mercury, drug and biocide resistance comprised 20% of resistance and detected only from urban ponds (Fig. 4B-D). High nitrate and nitrite concentrations and the presence of MRG were the main drivers of AMR abundance in the urban ponds. In contrast, high DO concentration and greater distance from the capital city separated rural ponds from the urban sites in terms of AMR abundance (Fig. 4E).

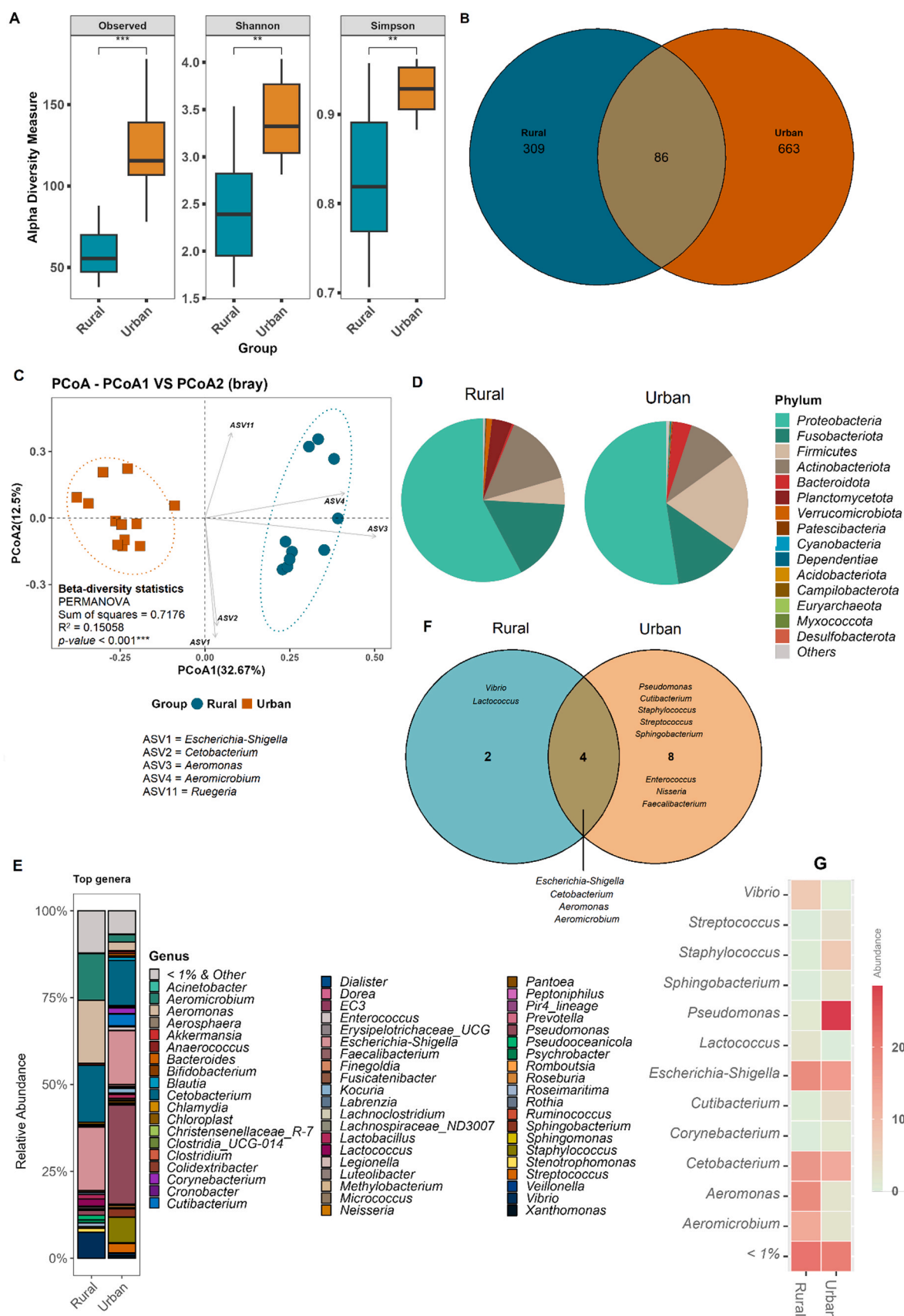


Fig. 1. Bacterial diversity and composition in rural and urban aquaculture ponds in Bangladesh. (A) Alpha-diversity in terms of species richness, Shannon and Simpson indices. (B) Number of shared and unique ASVs between two different groups. (C) Ordination based on “Bray-Curtis” dissimilarity of relative abundance. (D) Relative abundance of bacterial community at the phylum level. (E) Relative abundance of bacterial community at the genus level. (F) Core bacterial genera in two different groups (detected in >90% of samples). (G) Distribution of > 1% bacterial communities in two groups.

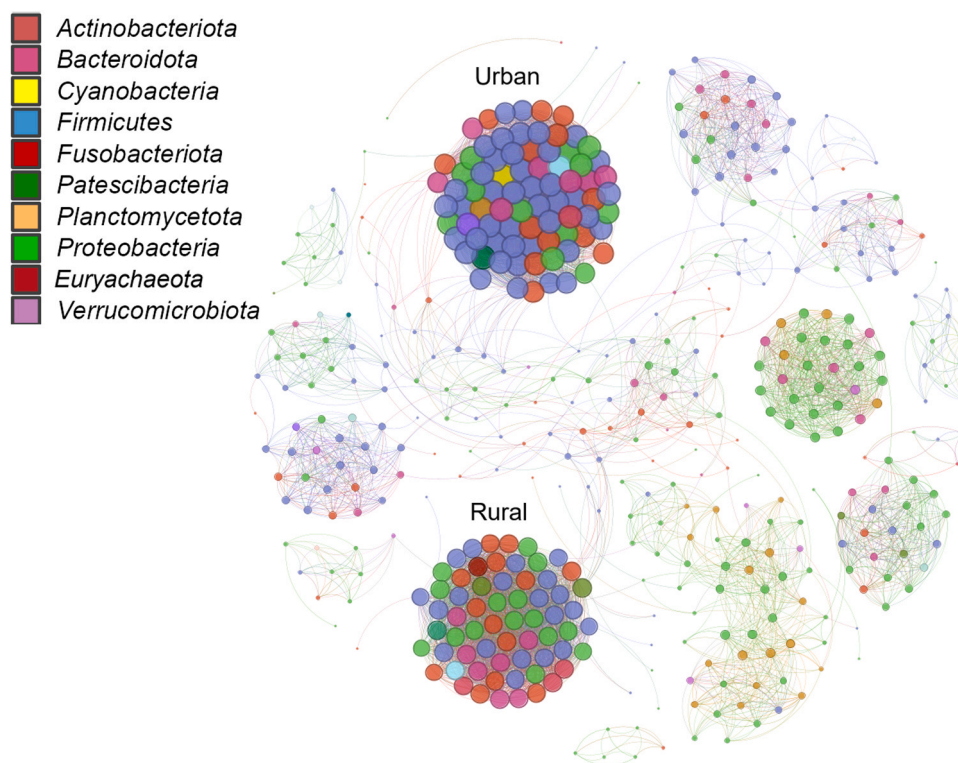


Fig. 2. Bacterial co-occurrence network between rural and urban pond samples. The color codes on the left represent specific phyla and the size of the nodes demonstrating the degree of interactions (1 = less interactive, 100 = highly interactive). The most interactive nodes for the rural and urban ponds are shown separately. The edges represent the distance between the nodes.

3.6. Microbiome-based AMR signature in the rural and urban ponds

High and medium-quality 35 bins were taxonomically classified against Kraken2 (Fig. S6). Alpha, Beta and Gamma-proteobacteria comprised 20 out of 35 bins while five were assigned to Cyanobacteria. Six bins, three from each of Alpha-proteobacteria and Cyanobacteria showed > 90% completeness and < 20% contamination (four had <10% contamination). Five other high-quality bins with 80–90% completeness and < 20% contamination were classified into Gamma-proteobacteria and one into Actinobacteria. The two complete (>99%) cyanobacterial bins were both classified into *Mycrocystis aeruginosa* (99% similarity). The other two alpha-proteobacterial bins were classified into Rhodobacterales (98% similarity) with lower resolution for *Rhodobacter* (82% similarity). Complete microcystin (*mcy*) gene clusters (*mcyA-J*) were identified for the bins classified into *M. aeruginosa*. No high or medium-quality bins were recovered for Firmicutes and therefore no bin-specific AMR was recovered from *Streptococcus*, *Staphylococcus* and *Enterococcus*. Beta and gamma-proteobacteria in urban ponds harbored 90% of the AMR and ESBL-genes. Two alphaproteobacterial and one gammaproteobacterial bins contained partial beta-lactamase gene clusters at 80–90% similarity (Fig. 5).

4. Discussion

We utilized both amplicon and metagenome-based sequencing to understand the microbial community structure and level of AMR in rural and urban ponds in Bangladesh. The urban pond water exhibited higher bacterial diversity, more AMR subclasses, and the presence of ESBL. The urban microbiome was predominantly driven by Firmicutes, including opportunistic pathogens such as *Streptococcus*, *Staphylococcus*, and *Enterococcus*. Overall, our results suggest the spread of AMR and opportunistic pathogens from clinical and industrial settings into urban aquaculture ponds, emphasizing the importance of proper waste management approaches in Bangladesh.

Apart from regional differences, the pond age, size, cultured species, and feeding ingredients were similar between urban and rural ponds, with no report of antibiotic use. External factors, particularly in urban areas, are more likely to significantly influence microbial community composition. Gazipur, a high-density area with major pharmaceutical and chemical industries, has elevated heavy metal concentrations, which can potentially impact the microbial community [66–70]. Additionally, the lack of automatic water removal facilities in this area can introduce MRG encoding bacteria from industrial run-off, particularly during the rainy season. This is specifically relevant given our study samples were collected during a period of heavy rainfall (July–August). Therefore, it is crucial to implement proper pond drainage systems with independent water outlets in urban areas. This will not only prevent the introduction of harmful bacteria but also ensure the safe and sanitary culture of aquatic species. Furthermore, regular monitoring and management of these factors can help to mitigate the risks associated with microbial imbalances in both urban and rural ponds and promote sustainable development.

The study reveals a uniform ($\geq 10\%$) distribution of *Cetobacterium* and *Escherichia-Shigella* across all samples, classifying them as ‘core’ water bacteria in polyculture ponds. This is irrespective of regional differences or other factors. The ponds used for farming three native Indian carp fish (rohu, catla and mrigal) known for their rich *Cetobacterium* presence [42,71], show a high abundance for these bacteria. The high abundance of *Cetobacterium* in the water correlates with the inextricable feces-water relationship in these polyculture ponds. However, the role of *Escherichia-Shigella* is unclear, hinting at potential human waste contamination. This is backed by the presence of *Enterococcus* in both rural and urban ponds and *Faecalibacterium* in urban ponds. Poor sanitation or sewer system leakages in both areas could lead to pond water contamination by human waste.

This study found a high abundance of Firmicutes in urban ponds. These bacteria, known for their diverse role in water ecosystems and resilience in antibiotic-rich environments [72,73], can be both

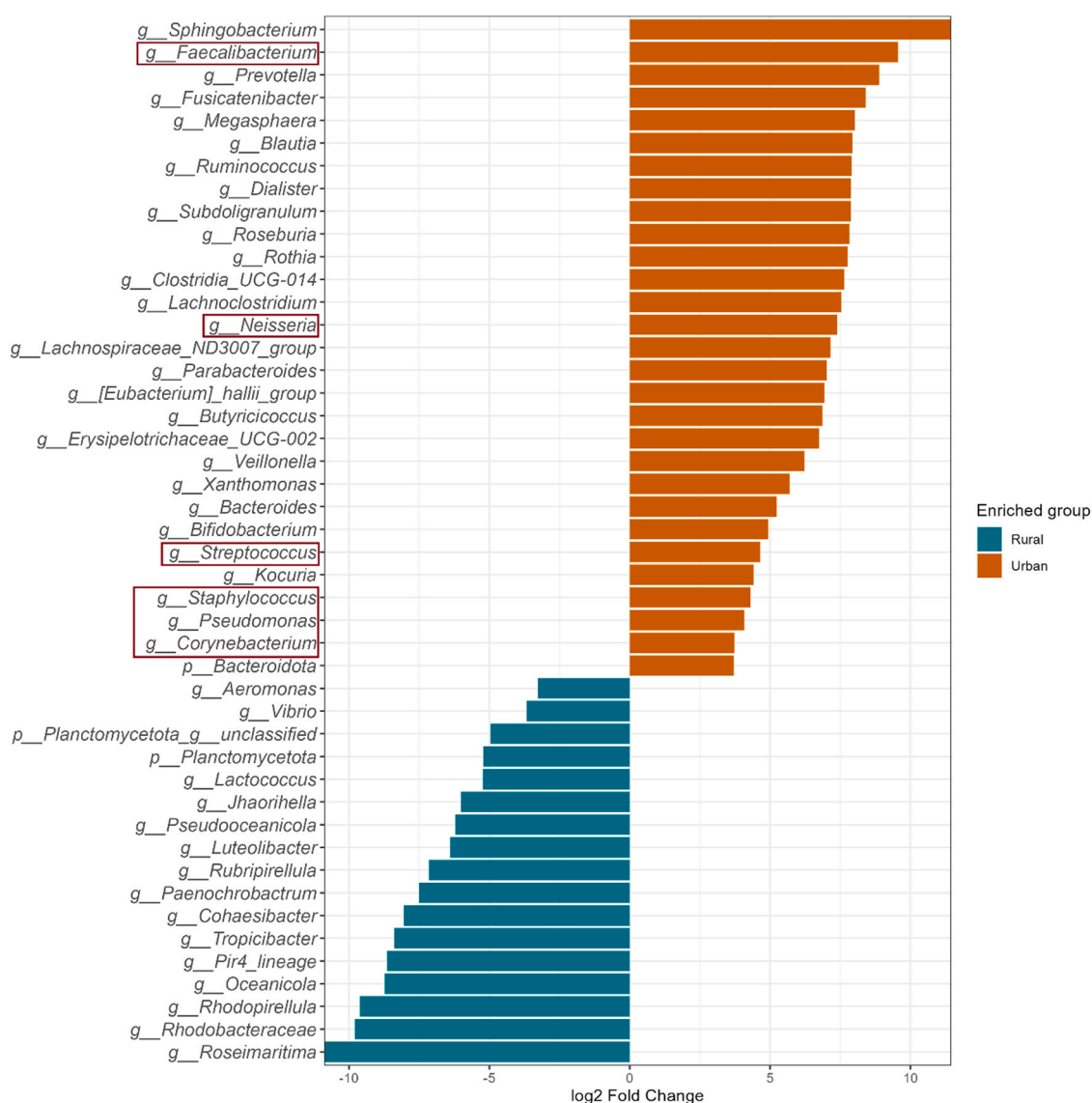


Fig. 3. Differential abundance of bacterial communities at genus level between rural and urban ponds. Genera with > 1% relative abundance were considered for differential abundance analysis. Opportunistic pathogens are marked with red color.

beneficial and harmful for humans and animals. Beneficial genera like *Bacillus*, *Lactobacillus*, and *Clostridium* support aquatic life [74,75], while *Streptococcus* and *Staphylococcus* can be opportunistic pathogens [76, 77]. *Enterococcus* can play dual roles depending on the habitat [78,79]. Firmicutes can form bacterial networks in harsh conditions to maintain key metabolic pathways [80,81]. Consistent with a previous study on AMR in semi-rural ponds [15], Firmicutes were less abundant in rural ponds. Therefore, we theorize that the robust network of Firmicutes in urban aquaculture ponds boosts their abundance in an AMR-rich environment, potentially serving as an AMR source.

This study differentiates rural and urban ponds based on the types of MRG in the water, aligning with research showing AMR spread in urban areas from industrial settings [66,82,83]. Notably, resistance to arsenic, copper, and mercury, ties to previous reports linking ARB with urban industrial effluents [27,66,84,82]. However, plate-based identification in these studies failed to directly link bacterial and metal resistance due to lack of metagenomic data. In contrast, our results support a study investigating the role of external contamination (like human waste), in shaping microbial communities and driving AMR prevalence [17]. Consistent with the findings of McInnes et al. [17], toxic bloom-forming *Cyanobium* and *Microcystis* were abundant in the metagenomic data. The

completed, contamination-free *Microcystis aeruginosa* bins contained full sets of toxic microcystin (*mcyA-mcyJ*) gene clusters [85,86]. Microcystin poses a serious health risk to aquatic and terrestrial organisms by affecting growth, metabolism, and immunity, creating a potential threat to animal health and environmental sustainability. In humans, even low level microcystin exposure can cause serious health disorders, including neurological disorders and cancer.

Heavy metals are prevalent in Bangladesh's water, yet reports on concentrations exceeding WHO limits are absent [24]. Additional subclasses of drug, biocide and metal (copper and mercury) resistance in urban ponds suggest pollution from chemical and industrial effluents. Two pollution drivers in urban ponds are the tannery industry's recent relocation from Dhaka to Gazipur, possibly contributing to metal resistance in the urban ponds [87] and insecticide use for mosquito control in major cities, potentially linked to biocide resistance [88,89,63]. Without heavy metal concentration measurements, the concern level is difficult to determine at this stage. However, the co-occurrence of ARG and metal-biocide resistance underscores the emerging health and environmental concerns of AMR in Bangladesh's urban aquaculture ponds.

The ESBL-producing clinical pathogens are a major concern in this study, mainly *Escherichia coli* and *Klebsiella pneumoniae*, *Salmonella*,

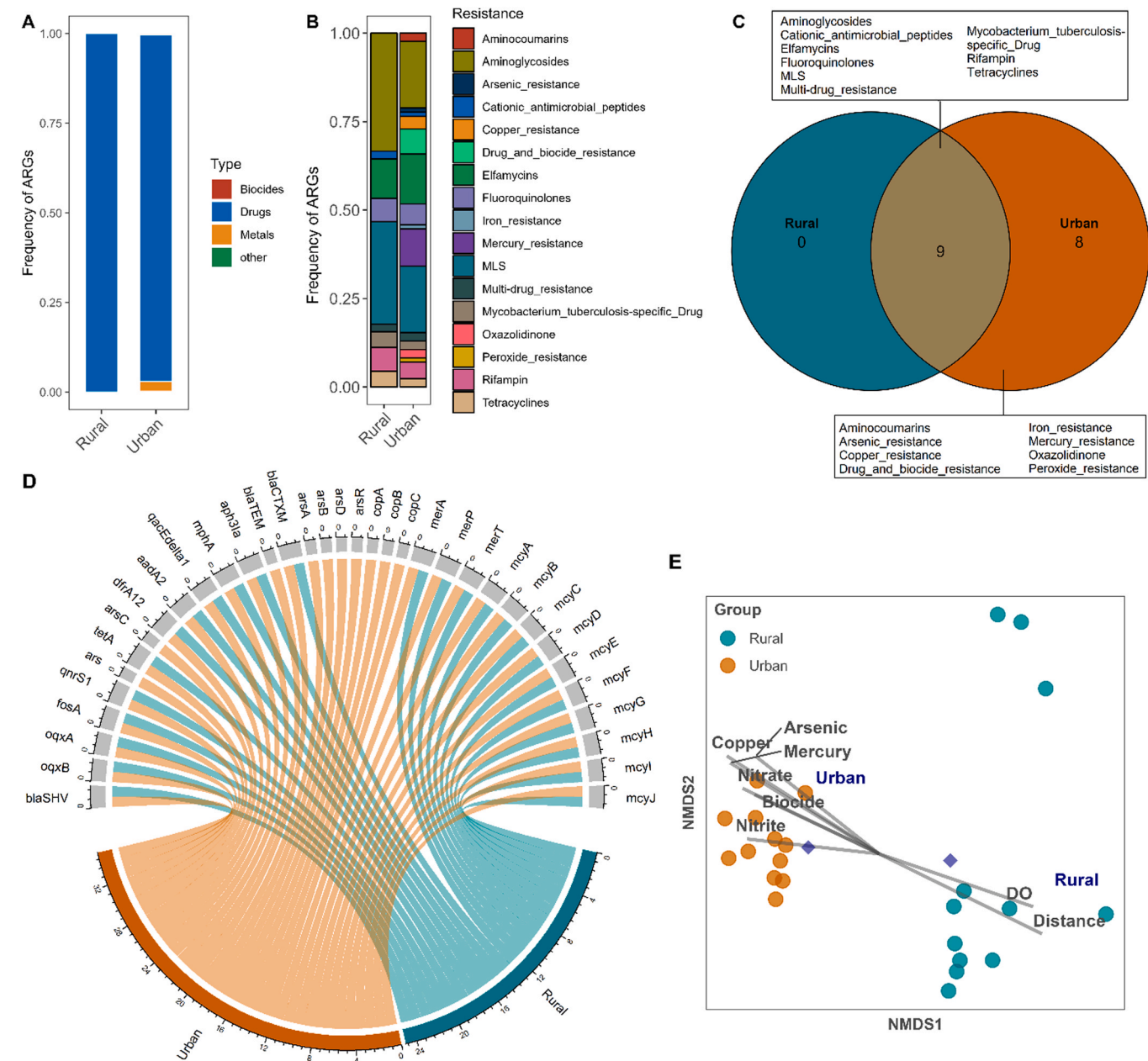


Fig. 4. Antimicrobial resistance in rural and urban ponds in Bangladesh. Types (A), subclasses (B) and their distributions (C) in the urban and rural ponds. (D) Circular plot showing the distribution of antibiotic and metal resistance and toxin genes in the rural and urban ponds based on diamond blast. (E) NMDS plot shows a correlation between bacterial community and environmental variables.

Shigella and *Vibrio*. The freshwater environment has been reported as the main source of ESBL-producing bacteria [90], including *Pseudomonas* [91], *E. coli* [92,93], and *Klebsiella pneumoniae* [94]. *Pseudomonas*, a potential ESBL-producer, requires further investigation considering its high abundance in urban ponds. The identified SHV and TEM ESBL-types are mostly associated with human infections such as diarrhea and urinary tract infections, signifying a major concern for human and environmental health in both rural and urban settings.

This study notes a disparity in the microbial community between the amplicon and metagenome datasets. While *Cyanobium*, *Limnohabitans*, and *Shewanella*, common in healthy aquaculture systems [95,96], were underrepresented or absent in the amplicon data, *Cutibacterium*, *Clostridium*, *Ruminococcus*, and *Rothia* were undetected in the metagenomic dataset. These variations could be due to the low resolution of short 16S rRNA Illumina reads and community variations in different hypervariable regions. Given the potential inconsistencies in amplicon data across

studies [97], combining both amplicon and metagenome-based sequencing can reduce biases in taxonomic and functional profiling.

5. Conclusions

The study reveals a high prevalence of ARB and AMR subclasses, including ESBL-producing genes, in urban pond water. Core water bacteria, *Escherichia-Shigella* and *Cetobacterium* are consistently abundant across all samples. However, the presence of opportunistic pathogens, and MRG in urban ponds raises significant food safety and health concerns. Despite the absence of MRG in rural ponds, the presence of opportunistic pathogens and cyanotoxins underscores the need for effective pond management and pollution awareness. Our study suggests that AMR spread can be mitigated by regulating environmental factors, given their positive correlation with bacterial communities. This study also emphasized the need for both amplicon and metagenome

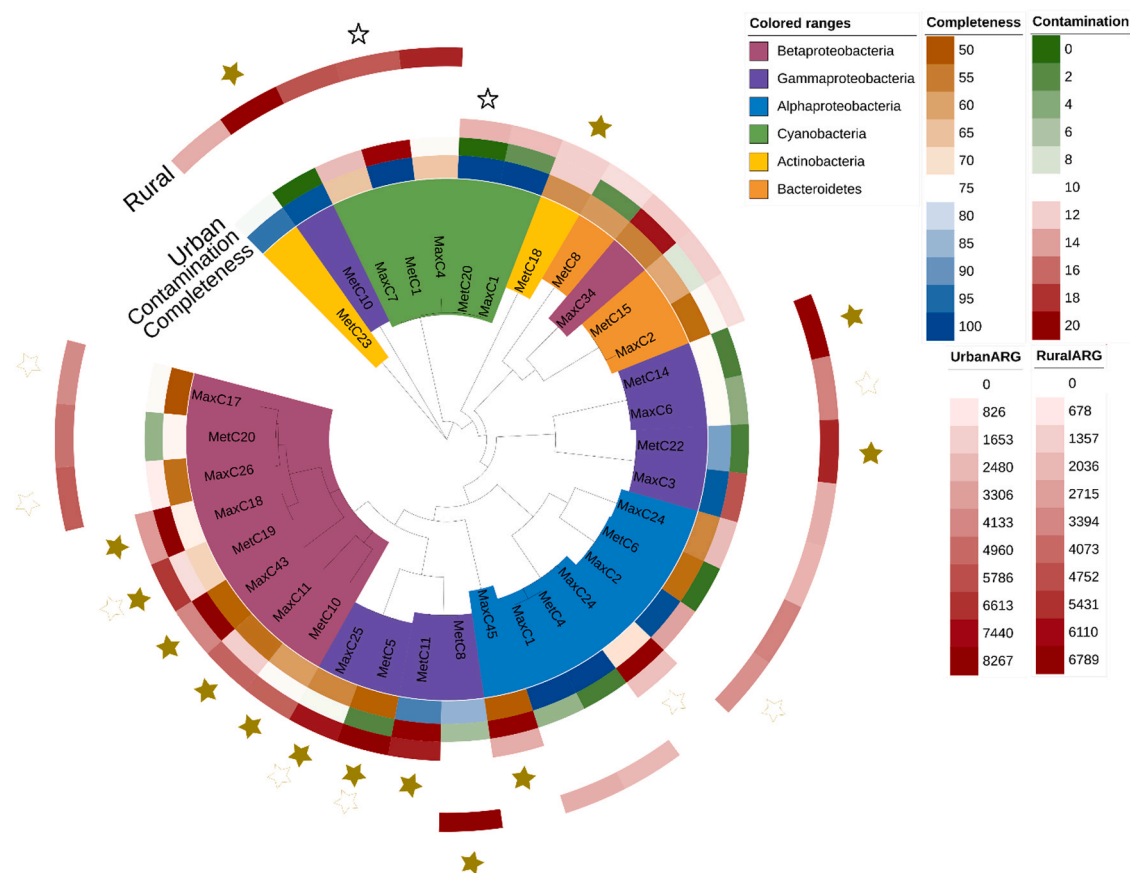


Fig. 5. Microbiome-based AMR signatures and toxin genes – characteristics of the metagenome-assembled genome (MAGs), AMR including MRG hits and beta-lactamase distributions in the high and medium quality bins. Filled gold stars represent the presence of beta-lactamase, and unfilled gold and black stars display positive bins for metal resistance and a full set of microcystin (*mcy*) gene clusters. The tree was created using GTOTree and visualized in iTOL.

sequencing to study microbial communities and AMR due to observed variations in microbial diversity and composition. Overall, this study provides crucial data for assessing the ecological and human risks of AMR, including MRG, in urban and rural ponds, thereby enhancing our understanding of their environmental mobility and promoting sustainable development in the blue economy.

Environmental implications

Antimicrobial resistance (AMR) and microbial metal resistance pose constant threats to environmental sustainability and human health. Aquaculture is one of the major protein sources for humans. This research investigates the microbial communities, AMR, and metal resistance in rural and urban ponds in Bangladesh. The goal is to understand how industrial pollution and anthropogenic activities impact the mobility of microbes, AMR, metal resistance, and toxin genes. The study aims to identify and evaluate the potential environmental and human health risks and the necessary steps to mitigate environmental threats and thereby promote the sustainable development of the blue economy.

CRedit authorship contribution statement

Md Javed Foysal: Writing – original draft, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Rabiul Kawser AQM:** Writing – original draft, Methodology, Investigation, Data curation. **Sulav Indra Paul:** Methodology, Data curation. **Md Reaz Chaklader:** Writing – original draft, Investigation. **Sanjay Kumar Gupta:** Writing – original draft, Investigation, Data curation. **Alfred Tay:** Resources, Methodology,

Investigation. **Brett Neilan:** Writing – review & editing, Supervision, Conceptualization. **Marthe Monique Gagnon:** Writing – review & editing. **Ravi Fotedar:** Writing – review & editing. **Md Mahbubur Rahman:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Verlaine Timms:** Writing – review & editing, Validation, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data used in this study are available in the 'Data availability' statement.

Acknowledgment

This research was crowd-funded (F-Project-3) by the research platform "Initiative for Decoding Research in Environmental and Aquatic Microbiome (iDREAM)". The authors would like to thank "The Marshall Centre", University of Western Australia for sequencing support. VT was funded by the Australian Research Council, Centre of Excellence in Synthetic Biology, Grant Number: CE200100029.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jhazmat.2024.134661](https://doi.org/10.1016/j.jhazmat.2024.134661).

References

- [1] Boyd, C.E., McNevin, A.A., Davis, R.P., 2022. The contribution of fisheries and aquaculture to the global protein supply. *Food Secur* 14 (3), 805–827. <https://doi.org/10.1007/s12571-021-01246-9>.
- [2] Dey, M.M., Alam, M.F., Bose, M.L., 2010. Demand for aquaculture development: perspectives from Bangladesh for improved planning. *Rev Aquac* 2 (1), 16–32. <https://doi.org/10.1111/j.1753-5131.2010.01020.x>.
- [3] Shamsuzzaman, M.M., Islam, M.M., Tania, N.J., Al-Mamun, M.A., Barman, P.P., Xu, X., 2017. Fisheries resources of Bangladesh: present status and future direction. *Aquacult Fish* 2 (4), 145–156. <https://doi.org/10.1016/j.aaf.2017.03.006>.
- [4] Rossignoli, C.M., Lozano Lazo, D.P., Barman, B.K., Dompheh, E.B., Manyise, T., Wang, Q., et al., 2023. Multi-stakeholder perception analysis of the status, characteristics, and factors affecting small-scale carp aquaculture systems in Bangladesh. *Front Sustain Food Syst* 7, 1121434. <https://doi.org/10.3389/fsufs.2023.1121434>.
- [5] Rana, S., Hasan, M.N., Al Bari, A., Shimul, S.A., Ahmed, S.I., Al Nahid, S.A., 2022. Problems and prospects of fish farming in the Chattogram Hill tracts of Bangladesh: community-based aquaculture might be a right choice. *Aquacult Fish* 9, 306–313. <https://doi.org/10.1016/j.aaf.2022.04.002>.
- [6] Hou, D., Zhou, R., Zeng, S., Wei, D., Deng, X., Xing, C., et al., 2021. Stochastic processes shape the bacterial community assembly in shrimp cultural pond sediments. *Appl Microbiol Biotechnol* 105 (12), 5013–5022. <https://doi.org/10.1007/s00253-021-11378-9>.
- [7] Rurangwa, E., Verdegem, M.C., 2015. Microorganisms in recirculating aquaculture systems and their management. *Rev Aquac* 7 (2), 117–130. <https://doi.org/10.1111/raq.12057>.
- [8] Meng, S., Peng, T., Liu, X., Wang, H., Huang, T., Gu, J.-D., et al., 2022. Ecological role of bacteria involved in the biogeochemical cycles of mangroves based on functional genes detected through GeoChip 5.0. *Mosphere* 7 (1), e00936–00921. <https://doi.org/10.1128/msphere.00936-21>.
- [9] Berg, G., Rybakova, D., Fischer, D., Cernava, T., Vergès, M.-C.C., Charles, T., et al., 2020. Microbiome definition re-visited: old concepts and new challenges. *Microbiome* 8 (1), 22. <https://doi.org/10.1186/s40168-020-00875-0>.
- [10] Xiao, G., Cheng, X., Zhu, D., Li, Z., Feng, L., Peng, X., et al., 2023. Exploring the mechanism of a novel recirculating aquaculture system based on water quality parameters and bacterial communities. *Environ Sci Pollut Res* 30 (12), 34760–34774. <https://doi.org/10.1007/s11356-022-24585-2>.
- [11] Zhang, M.-Q., Yang, J.-L., Lai, X.-X., Li, W., Zhan, M.-J., Zhang, C.-P., et al., 2022. Effects of integrated multi-trophic aquaculture on microbial communities, antibiotic resistance genes, and cultured species: a case study of four mariculture systems. *Aquaculture* 557, 738322. <https://doi.org/10.1016/j.aquaculture.2022.738322>.
- [12] Hassan, M.A., Fathallah, M.A., Elzoghby, M.A., Salem, M.G., Helmy, M.S., 2022. Influence of probiotics on water quality in intensified *Litopenaeus vannamei* ponds under minimum-water exchange. *AMB Express* 12 (1), 22. <https://doi.org/10.1186/s13568-022-01370-5>.
- [13] Foyzal, M.J., Fotedar, R., Tay, C.-Y., Gupta, S.K., 2020. Biological filters regulate water quality, modulate health status, immune indices and gut microbiota of freshwater crayfish, *marron (Cherax cainii)*, Austin, 2002). *Chemosphere* 247, 125821. <https://doi.org/10.1016/j.chemosphere.2020.125821>.
- [14] Martinez-Cordova, L.R., Emerenciano, M.G., Miranda-Baeza, A., Pinho, S.M., Garibay-Valdez, E., Martinez-Porchas, M., 2023. Advancing toward a more integrated aquaculture with polyculture > aquaponics > biofloc technology > FLOCponics. *Aquac Int* 31 (2), 1057–1076. <https://doi.org/10.1007/s10499-022-01016-0>.
- [15] Bell, A.G., Thornber, K., Chaput, D.L., Hasan, N.A., Alam, M.M., Haque, M.M., et al., 2023. Metagenomic assessment of the diversity and ubiquity of antimicrobial resistance genes in Bangladeshi aquaculture ponds. *Aquacult Rep* 29, 101462. <https://doi.org/10.1016/j.aqrep.2023.101462>.
- [16] Lassen, S.B., Ahsan, M.E., Islam, S.R., Zhou, X.-Y., Razzak, M.A., Su, J.-Q., et al., 2022. Prevalence of antibiotic resistance genes in Pangasianodon hypophthalmus and Oreochromis niloticus aquaculture production systems in Bangladesh. *Sci Total Environ* 813, 151915. <https://doi.org/10.1016/j.scitotenv.2021.151915>.
- [17] McInnes, R.S., Uz-Zaman, M.H., Alam, I.T., Ho, S.F.S., Moran, R.A., Clemens, J.D., et al., 2021. Metagenome-wide analysis of rural and urban surface waters and sediments in Bangladesh identifies human waste as a driver of antibiotic resistance. *Msystems* 6 (4), e00137–00121. <https://doi.org/10.1128/msystems.00137-21>.
- [18] Rashid, M., Rakib, M.M., Hasan, B., 2015. Antimicrobial-resistant and ESBL-producing *Escherichia coli* in different ecological niches in Bangladesh. *Infect Ecol Epidemiol* 5 (1), 26712. <https://doi.org/10.3402/iee.v5.26712>.
- [19] Sarker, M.S., Ahad, A., Ghosh, S.K., Mannan, M.S., Sen, A., Islam, S., et al., 2019. Antibiotic-resistant *Escherichia coli* in deer and nearby water sources at Safari parks in Bangladesh. *Vet World* 12 (10), 1578. <https://doi.org/10.14202/vetworld.2019.1578-1583>.
- [20] Ma, F., Xu, S., Tang, Z., Li, Z., Zhang, L., 2021. Use of antimicrobials in food animals and impact of transmission of antimicrobial resistance on humans. *Biosaf Health* 3 (1), 32–38. <https://doi.org/10.1016/j.bshealth.2020.09.004>.
- [21] Ye, L., Liu, G., Yao, T., Lu, J., 2021. Monitoring of antimicrobial resistance genes in the spotted sea bass (*Lateolabrax maculatus*): Association with the microbiome and its environment in aquaculture ponds. *Environ Pollut* 276, 116714. <https://doi.org/10.1016/j.envpol.2021.116714>.
- [22] Azmuda, N., Fakruddin, M., Khan, S.I., Birkeland, N.-K., 2019. Bacterial community profiling of tropical freshwaters in Bangladesh. *Front Public Health* 7, 115. <https://doi.org/10.3389/fpubh.2019.00115>.
- [23] Foster, T., Falletta, J., Amin, N., Rahman, M., Liu, P., Raj, S., et al., 2021. Modelling faecal pathogen flows and health risks in urban Bangladesh: implications for sanitation decision making. *Int J Hyg Environ Health* 233, 113669. <https://doi.org/10.1016/j.ijheh.2020.113669>.
- [24] Hasan, M.K., Shahriar, A., Jim, K.U., 2019. Water pollution in Bangladesh and its impact on public health. *Heliyon* 5 (8). <https://doi.org/10.1016/j.heliyon.2019.e02145>.
- [25] Saima, S., Ferdous, J., Sultana, R., Rashid, R.B., Almeida, S., Begum, A., et al., 2023. Detecting enteric pathogens in low-risk drinking water in Dhaka, Bangladesh: An assessment of the WHO water safety categories. *Trop Med Infect Dis* 8 (6), 321. <https://doi.org/10.3390/tropicalmed8060321>.
- [26] Thornber, K., Bashir, A., Ahmed, M.S., Bell, A., Trew, J., Hasan, M., et al., 2022. Antimicrobial resistance in aquaculture environments: unravelling the complexity and connectivity of the underlying societal drivers. *Environ Sci Technol* 56 (21), 14891–14903. <https://doi.org/10.1021/acs.est.2c00799>.
- [27] Ahmad, I., Malak, H.A., Abulreesh, H.H., 2021. Environmental antimicrobial resistance and its drivers: a potential threat to public health. *J Glob Antimicrob Resist* 27, 101–111. <https://doi.org/10.1016/j.jgar.2021.08.001>.
- [28] Karkman, A., Do, T.T., Walsh, F., Virta, M.P., 2018. Antibiotic-resistance genes in waste water. *Trends Microbiol* 26 (3), 220–228. <https://doi.org/10.1016/j.tim.2017.09.005>.
- [29] Le, T., Truong, T., Tran, L.-T., Nguyen, D.-H., Pham, T., Ng, C., 2023. Antibiotic resistance in the aquatic environments: the need for an interdisciplinary approach. *Int J Environ Sci Technol (Tehran)* 20 (3), 3395–3408.
- [30] Zhu, Y.-G., Johnson, T.A., Su, J.-Q., Qiao, M., Guo, G.-X., Stedtfeld, R.D., et al., 2013. Diverse and abundant antibiotic resistance genes in Chinese swine farms. *PNAS* 110 (9), 3435–3440. <https://doi.org/10.1073/pnas.1222743110>.
- [31] Holden, J.F., Adams, M.W., 2003. Microbe–metal interactions in marine hydrothermal environments. *Curr Opin Chem Biol* 7 (2), 160–165. [https://doi.org/10.1016/S1367-5931\(03\)00026-7](https://doi.org/10.1016/S1367-5931(03)00026-7).
- [32] Olden, K., White, S.L., 2005. Health-related disparities: influence of environmental factors. *Med Clin North Am* 89 (4), 721–738. <https://doi.org/10.1016/j.mcna.2005.02.001>.
- [33] Buelow, E., Rico, A., Gaschet, M., Lourenço, J., Kennedy, S.P., Wiest, L., Ploy, M.-C., Dagot, C., 2020. Hospital discharges in urban sanitation systems: Long-term monitoring of wastewater resistome and microbiota in relationship to their eco-exposome. *Water Res* 177, 115845. <https://doi.org/10.1016/j.watres.2020.115845>.
- [34] Scholz, S., Nichols, J.W., Esche, B.I., Ankley, G.T., Altenburger, R., Blackwell, B., Brack, W., Burkhard, L., Collette, T.W., Doering, J.A., 2022. The Eco-Exposome concept: Supporting an integrated assessment of mixtures of environmental chemicals. *Environ Toxicol Chem* 41 (1), 30–45. <https://doi.org/10.1002/etc.5242>.
- [35] Kim, S.O., Kim, S.S., 2021. Bacterial pathogen detection by conventional culture-based and recent alternative (polymerase chain reaction, isothermal amplification, enzyme linked immunosorbent assay, bacteriophage amplification, and gold nanoparticle aggregation) methods in food samples: a review. *J Food Saf* 41 (1), e12870. <https://doi.org/10.1111/jfs.12870>.
- [36] Su, X., Bamba, A., Zhang, S., Zhang, Y., Hashmi, M., Lin, H., et al., 2018. Revealing potential functions of VBNC bacteria in polycyclic aromatic hydrocarbons biodegradation. *Lett Appl Microbiol* 66 (4), 277–283. <https://doi.org/10.1111/lam.12853>.
- [37] Craud, P., Rasplus, J.-Y., Rodriguez, L.J., Craud, A., 2017. High-throughput sequencing of multiple amplicons for barcoding and integrative taxonomy. *Sci Rep* 7 (1), 41948. <https://doi.org/10.1038/srep41948>.
- [38] Reuter, J.A., Spacek, D.V., Snyder, M.P., 2015. High-throughput sequencing technologies. *Mol Cell* 58 (4), 586–597. <https://doi.org/10.1016/j.molcel.2015.05.004>.
- [39] Hugerth, L.W., Andersson, A.F., 2017. Analysing microbial community composition through amplicon sequencing: from sampling to hypothesis testing. *Front Microbiol* 8, 1561. <https://doi.org/10.3389/fmicb.2017.01561>.
- [40] Poretsky, R., Rodriguez-R, L.M., Luo, C., Tsementzi, D., Konstantinidis, K.T., 2014. Strengths and limitations of 16S rRNA gene amplicon sequencing in revealing temporal microbial community dynamics. *PLoS One* 9 (4), e93827. <https://doi.org/10.1371/journal.pone.0093827>.
- [41] Campanaro, S., Treu, L., Kougiou, P.G., Luo, G., Angelidaki, I., 2018. Metagenomic binning reveals the functional roles of core abundant microorganisms in twelve full-scale biogas plants. *Water Res* 140, 123–134. <https://doi.org/10.1016/j.watres.2018.04.043>.
- [42] Foyzal, M.J., Momtaz, F., Kawsar, A., Rahman, M., Gupta, S.K., Tay, A.C.Y., 2020. Next-generation sequencing reveals significant variations in bacterial compositions across the gastrointestinal tracts of the Indian major carps, rohu (*Labeo rohita*), catla (*Catla catla*) and mrigal (*Cirrhinus cirrhosis*). *Lett Appl Microbiol* 70 (3), 173–180. <https://doi.org/10.1111/lam.13256>.
- [43] Klindworth, A., Pruesse, E., Schweer, T., Piephs, J., Quast, C., Horn, M., et al., 2013. Evaluation of general 16S ribosomal RNA gene PCR primers for classical and next-generation sequencing-based diversity studies. *Nucleic Acids Res* 41 (1) e1–e1. <https://doi.org/10.1093/nar/gks808>.
- [44] Bolyen, E., Rideout, J.R., Dillon, M.R., Bokulich, N.A., Abnet, C.C., Al-Ghalith, G. A., et al., 2019. Reproducible, interactive, scalable and extensible microbiome data

- science using QIIME 2. *Nat Biotechnol* 37 (8), 852–857. <https://doi.org/10.1038/s41587-019-0209-9>.
- [45] Callahan, B.J., McMurdie, P.J., Rosen, M.J., Han, A.W., Johnson, A.J.A., Holmes, S. P., 2016. DADA2: high-resolution sample inference from Illumina amplicon data. *Nat Methods* 13 (7), 581–583. <https://doi.org/10.1038/nmeth.3869>.
- [46] Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., et al., 2012. The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. *Nucleic Acids Res* 41 (D1), D590–D596. <https://doi.org/10.1093/nar/gks1219>.
- [47] Allaire, J., Ellis, P., Gandrud, C., Kuo, K., Lewis, B., Owen, J. et al., 2017. Package 'networkD3'. D3 JavaScript network graphs from R.
- [48] Csardi, G., Nepusz, T., 2006. The igraph software package for complex network research. *Inter. Complex Syst* 1695 (5), 1–9.
- [49] Bastian, M., Heymann, S., Jacomy, M., 2009. Gephi: an open source software for exploring and manipulating networks. *Proc Int AAAI Conf web Soc Media* 361–362.
- [50] Bushnell, B., 2014. BBMap: a fast, accurate, splice-aware aligner. Lawrence Berkeley National Lab.(LBNL), Berkeley, CA (United States).
- [51] Andrews, S., 2010. FastQC: a quality control tool for high throughput sequence data [Software]. p. Available (<http://www.bioinformatics.babraham.ac.uk/project/fastqc/>).
- [52] Li, D., Liu, C.-M., Luo, R., Sadakane, K., Lam, T.-W., 2015. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. *Bioinformatics* 31 (10), 1674–1676. <https://doi.org/10.1093/bioinformatics/btv033>.
- [53] Langmead, B., Salzberg, S.L., 2012. Fast gapped-read alignment with Bowtie2. *Nat Methods* 9 (4), 357–359. <https://doi.org/10.1038/nmeth.1923>.
- [54] Li, H., Handsaker, B., Wysoker, A., Fennell, T., Ruan, J., Homer, N., et al., 2009. The sequence alignment/map format and SAMtools. *Bioinformatics* 25 (16), 2078–2079. <https://doi.org/10.1093/bioinformatics/btp352>.
- [55] Kang, D.D., Li, F., Kirtan, E., Thomas, A., Egan, R., An, H., et al., 2019. MetaBAT2: an adaptive binning algorithm for robust and efficient genome reconstruction from metagenome assemblies. *PeerJ* 7, e7359. <https://doi.org/10.7717/peerj.7359>.
- [56] Wu, Y.-W., Simmons, B.A., Singer, S.W., 2016. MaxBin 2.0: an automated binning algorithm to recover genomes from multiple metagenomic datasets. *Bioinformatics* 32 (4), 605–607. <https://doi.org/10.1093/bioinformatics/btv638>.
- [57] Nayfach, S., Shi, Z.J., Seshadri, R., Pollard, K.S., Kyrpides, N.C., 2019. New insights from uncultivated genomes of the global human gut microbiome. *Nature* 568 (7753), 505–510. <https://doi.org/10.1038/s41586-019-1058-x>.
- [58] Sieber, C.M., Probst, A.J., Sharrar, A., Thomas, B.C., Hess, M., Tringe, S.G., et al., 2018. Recovery of genomes from metagenomes via a dereplication, aggregation and scoring strategy. *Nat Microbiol* 3 (7), 836–843. <https://doi.org/10.1038/s41564-018-0171-1>.
- [59] Parks, D.H., Imelfort, M., Skennerton, C.T., Hugenholtz, P., Tyson, G.W., 2015. CheckM: assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes. *Genome Res* 25 (7), 1043–1055. <https://doi.org/http://www.genome.org/cgi/doi/10.1101/gr.186072.114>.
- [60] Wood, D.E., Lu, J., Langmead, B., 2019. Improved metagenomic analysis with Kraken 2. *Genome Biol* 20 (1), 13. <https://doi.org/10.1186/s13059-019-1891-0>.
- [61] Bonin, N., Doster, E., Worley, H., Pinnell, L.J., Bravo, J.E., Ferm, P., et al., 2023. MEGARes and AMR++, v3.0: an updated comprehensive database of antimicrobial resistance determinants and an improved software pipeline for identification using high-throughput sequencing. *Nucleic Acids Res* 51 (D1), D744–D752. <https://doi.org/10.1093/nar/gkac1047>.
- [62] Sherry, N.L., Horan, K.A., Ballard, S.A., Gonçalves da Silva, A., Gorrie, C.L., Schultz, M.B., et al., 2023. An ISO-certified genomics workflow for identification and surveillance of antimicrobial resistance. *Nat Com* 14 (1), 60. <https://doi.org/10.1038/s41467-022-35713-4>.
- [63] Buchfink, B., Reuter, K., Drost, H.-G., 2021. Sensitive protein alignments at tree-of-life scale using DIAMOND. *Nat Methods* 18 (4), 366–368. <https://doi.org/10.1038/s41467-022-35713-4>.
- [64] Letunic, I., Bork, P., 2021. Interactive Tree Of Life (iTOL) v5: an online tool for phylogenetic tree display and annotation. *Nucleic Acids Res* 49 (W1), W293–W296. <https://doi.org/10.1093/nar/gkab301>.
- [65] Cao, Y., Dong, Q., Wang, D., Zhang, P., Liu, Y., Niu, C., 2022. microbiomeMarker: an R/Bioconductor package for microbiome marker identification and visualization. *Bioinformatics* 38 (16), 4027–4029. <https://doi.org/10.1093/bioinformatics/btac438>.
- [66] Ahmed, M., Matsumoto, M., Kurosawa, K., 2018. Heavy metal contamination of irrigation water, soil, and vegetables in a multi-industry district of Bangladesh. *Int J Environ Res* 12, 531–542. <https://doi.org/10.3390/w11030583>.
- [67] Ahmed, M., Matsumoto, M., Ozaki, A., Thinh, N.V., Kurosawa, K., 2019. Heavy metal contamination of irrigation water, soil, and vegetables and the difference between dry and wet seasons near a multi-industry zone in Bangladesh. *Water* 11 (3), 583. <https://doi.org/10.3390/w11030583>.
- [68] Hafizur, R., Nuralam, H., Rumainul, I., 2017. Investigation of physicochemical parameter, heavy metal in Turag river water and adjacent industrial effluent in Bangladesh. *J Sci Technol Environ Info* 5 (1), 347–360. <https://doi.org/10.18801/jstei.050117.37>.
- [69] Hasan, G., Das, A.K., Satter, M.A., 2022. Accumulation of heavy metals in rice (*Oryza sativa*. L) grains cultivated in three major industrial areas of Bangladesh. *J Environ Public Health* 2022. <https://doi.org/10.1155/2022/1836597>.
- [70] Islam, M., Tusher, T., Mustafa, M., Mahmud, S., 2012. Effects of solid waste and industrial effluents on water quality of Turag River at Konabari industrial area, Gazipur, Bangladesh. *J Environ Sci Nat Resour* 5 (2), 213–218. <https://doi.org/10.3329/jesnr.v5i2.14817>.
- [71] Mukherjee, A., Rodiles, A., Merrifield, D.L., Chandra, G., Ghosh, K., 2020. Exploring intestinal microbiome composition in three Indian major carps under polyculture system: a high-throughput sequencing based approach. *Aquaculture* 524, 735206. <https://doi.org/10.1016/j.aquaculture.2020.735206>.
- [72] Guan, Y., Jia, J., Wu, L., Xue, X., Zhang, G., Wang, Z., 2018. Analysis of bacterial community characteristics, abundance of antibiotics and antibiotic resistance genes along a pollution gradient of Ba River in Xi'an, China. *Front Microbiol* 9, 3191. <https://doi.org/10.3389/fmicb.2018.03191>.
- [73] Novo, A., André, S., Viana, P., Nunes, O.C., Manaia, C.M., 2013. Antibiotic resistance, antimicrobial residues and bacterial community composition in urban wastewater. *Water Res* 47 (5), 1875–1887. <https://doi.org/10.1016/j.watres.2013.01.010>.
- [74] Foytal, M.J., Dao, T.T.T., Fotedar, R., Gupta, S.K., Tay, A., Chaklader, M.R., 2022. Sources of protein diet differentially stimulate the gut and water microbiota under freshwater crayfish, marron (*Cherax cainii*, Austin 2002) culture. *Environ Microbiol Rep* 14 (2), 286–298. <https://doi.org/10.1111/1758-2229.13049>.
- [75] Kuebutornye, F.K., Abarike, E.D., Lu, Y., Hlordzi, V., Sakyi, M.E., Afriyie, G., et al., 2020. Mechanisms and the role of probiotic *Bacillus* in mitigating fish pathogens in aquaculture. *Fish Physiol Biochem* 46, 819–841. <https://doi.org/10.1007/s10695-019-00754-y>.
- [76] Itay, P., Shemesh, E., Ofek-Lazlar, M., Davidovich, N., Kroin, Y., Zrihan, S., et al., 2022. An insight into gill microbiome of Eastern Mediterranean wild fish by applying next generation sequencing. *Front Mar Sci* 9, 1008103. <https://doi.org/10.3389/fmars.2022.1008103>.
- [77] Spaulding, A.R., Salgado-Pabón, W., Kohler, P.L., Horswill, A.R., Leung, D.Y., Schlievert, P.M., 2013. Staphylococcal and streptococcal superantigen exotoxins. *Clin Microbiol Rev* 26 (3), 422–447. <https://doi.org/10.1128/CMR.00104-12>.
- [78] Adnan, M., Patel, M., Hadi, S., 2017. Functional and health promoting inherent attributes of *Enterococcus hirae* F2 as a novel probiotic isolated from the digestive tract of the freshwater fish *Catla catla*. *PeerJ* 5, e3085. <https://doi.org/10.7717/peerj.3085>.
- [79] Akter, T., Haque, M.N., Ehsan, R., Paul, S.I., Foytal, M.J., Tay, A.C.Y., et al., 2023. Virulence and antibiotic-resistance genes in *Enterococcus faecalis* associated with streptococcosis disease in fish. *Sci Rep* 13 (1), 1551. <https://doi.org/10.1038/s41598-022-25968-8>.
- [80] Borenstein, E., Feldman, M.W., 2009. Topological signatures of species interactions in metabolic networks. *J Comput Biol* 16 (2), 191–200. <https://doi.org/10.1089/cmb.2008.06TT>.
- [81] Zhou, H., Gao, Y., Jia, X., Wang, M., Ding, J., Cheng, L., et al., 2020. Network analysis reveals the strengthening of microbial interaction in biological soil crust development in the Mu Us Sandy Land, northwestern China. *Soil Biol Biochem* 144, 107782. <https://doi.org/10.1016/j.soilbio.2020.107782>.
- [82] Marzan, L.W., Hossain, M., Mina, S.A., Akter, Y., Chowdhury, A.M.A., 2017. Isolation and biochemical characterization of heavy-metal resistant bacteria from tannery effluent in Chittagong city, Bangladesh: Bioremediation viewpoint. *Egypt J Aquat Res* 43 (1), 65–74. <https://doi.org/10.1016/j.ejar.2016.11.002>.
- [83] Rahman, M.H., Ferdouse, J., Ullah, A.A., Hossain, M.T., 2022. Water quality assessment and identification of novel bacterial strains in the Halda river water of Bangladesh. *Air, Soil Wat Res* 15, 1–10. <https://doi.org/10.1177/11786221221135164>.
- [84] Danko, D., Bezdán, D., Afshin, E.E., Ahsanuddin, S., Bhattacharya, C., Butler, D.J., et al., 2021. A global metagenomic map of urban microbiomes and antimicrobial resistance. *e3317 Cell* 184 (13), 3376–3393. <https://doi.org/10.1016/j.cell.2021.05.002>.
- [85] Lad, A., Breidenbach, J.D., Su, R.C., Murray, J., Kuang, R., Mascarenhas, A., et al., 2022. As we drink and breathe: adverse health effects of microcystins and other harmful algal bloom toxins in the liver, gut, lungs and beyond. *Life* 12 (3), 418. <https://doi.org/10.3390/life12030418>.
- [86] Zanchett, G., Oliveira-Filho, E.C., 2013. Cyanobacteria and cyanotoxins: from impacts on aquatic ecosystems and human health to anticarcinogenic effects. *Toxins (Basel)* 5 (10), 1896–1917. <https://doi.org/10.3390/toxins5101896>.
- [87] Shaibur, M.R., 2023. Heavy metals in chrome-tanned shaving of the tannery industry are a potential hazard to the environment of Bangladesh. *Case Stud Chem Environ Engg* 7, 100281. <https://doi.org/10.1016/j.cscee.2022.100281>.
- [88] Al-Amin, H.M., Gyawali, N., Graham, M., Alam, M.S., Lenhart, A., Hugo, L.E., et al., 2023. Insecticide resistance compromises the control of *Aedes aegypti* in Bangladesh. *Pest Manag Sci* 79, 2846–2861. <https://doi.org/10.1002/ps.7462>.
- [89] Bhute, S.S., Escobedo, B., Haider, M., Mekonen, Y., Ferrer, D., Hillyard, S.D., et al., 2020. The gut microbiome and its potential role in paradoxical anaerobism in pupfishes of the Mojave Desert. *Anim Micro* 2 (1), 1–9. <https://doi.org/10.1186/s42523-020-00037-5>.
- [90] Cho, S., Jackson, C.R., Frye, J.G., 2023. Freshwater environment as a reservoir of extended-spectrum β -lactamase-producing Enterobacteriaceae. *J Appl Microbiol* 134 (3), lxad034. <https://doi.org/10.1093/jambio/lxad034>.
- [91] Falodun, O.I., Ikusika, E.O., 2020. Extended spectrum beta-lactamase and metallo beta-lactamase producing *Pseudomonas* species isolated from fish pond water in Ibadan, Nigeria. *Int J Environ Stud* 77 (5), 865–875. <https://doi.org/10.1080/00207233.2019.1705044>.
- [92] Hoa, T., Nakayama, T., Huyen, H., Harada, K., Hinenoya, A., Phuong, N., et al., 2020. Extended-spectrum beta-lactamase-producing *Escherichia coli* harbouring sul and mcr-1 genes isolates from fish gut contents in the Mekong Delta, Vietnam. *Lett Appl Microbiol* 71 (1), 78–85. <https://doi.org/10.1111/lam.13222>.
- [93] Tran, H.L., Hong, M.H., Tran, T.T.H., 2018. Antibiotic resistance and molecular characteristics of extended-spectrum beta-lactamase-producing *Escherichia coli* isolated from fish pond. *Can Tho Uni J Sci* 54(8 114–123. <https://doi.org/10.22144/ctu.jen.2018.045>.

- [94] Sivaraman, G.K., Rajan, V., Vijayan, A., Elangovan, R., Prendiville, A., Bachmann, T.T., 2021. Antibiotic resistance profiles and molecular characteristics of extended-spectrum beta-lactamase (ESBL)-producing *Escherichia coli* and *Klebsiella pneumoniae* isolated from shrimp aquaculture farms in Kerala, India. *Front Microbiol* 12, 622891. <https://doi.org/10.3389/fmicb.2021.622891>.
- [95] Huang, Z., Jiang, C., Xu, S., Zheng, X., Lv, P., Wang, C., et al., 2022. Spatiotemporal changes of bacterial communities during a cyanobacterial bloom in a subtropical water source reservoir ecosystem in China. *Sci Rep* 12 (1), 14573. <https://doi.org/10.1038/s41598-022-17788-7>.
- [96] Savvichev, A.S., Kadnikov, V.V., Rusanov, I.I., Beletsky, A.V., Krasnova, E.D., Voronov, D.A., et al., 2020. Microbial processes and microbial communities in the water column of the polar meromictic Lake Bol'shie Khruslomeny at the White Sea Coast. *Front Microbiol* 11, 1945. <https://doi.org/10.3389/fmicb.2020.01945>.
- [97] Rausch, P., Rühlemann, M., Hermes, B.M., Doms, S., Dagan, T., Dierking, K., et al., 2019. Comparative analysis of amplicon and metagenomic sequencing methods reveals key features in the evolution of animal metaorganisms. *Microbiome* 7 (1), 1–19. <https://doi.org/10.1186/s40168-019-0743-1>.